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ECA5604-Communication Systems

CO1 Integrated Engineering Analysis and Planning Report

Evolution, cellular planning, spectrum use and interference management for a regional mobile operator

Executive summary

This report analyses the supplied planning data and recommends a staged network plan. The key results are:

- Offered busy-hour traffic is A = 80 Erlang. With 100 channels, Erlang B blocking is 0.40 %; the smallest channel count meeting the 2 % target is 92 channels (1.92 %). One 100-channel group is therefore enough in aggregate, but a realistic 19-cell layout needs about 171 channels (9 per cell) because traffic cannot be pooled across cells.
- Hexagonal cell area is 10.39 km², the 19-cell cluster covers 197.45 km², the reuse fraction for N = 7 is 1/7, and the reuse distance is D = 9.17 km (D/R = 4.58).
- At 60 km/h the mobile covers 100 m in the 6 s overlap zone and makes about 1 hard handoff per minute; the simple call-drop risk is about 1 % per handoff.
- The interference case gives C/I = 15.0 dB from the co-channel signal alone, but total SINR = 13.81 dB, which fails the 15 dB target by 1.19 dB. Sectorization is recommended.
- Free-space path loss at 100 m is 71.5 dB (900 MHz), 83.3 dB (3.5 GHz) and 101.4 dB (28 GHz). The recommended strategy is a three-phase plan: sectorize and densify on 900 MHz / sub-GHz coverage first, add 3.5 GHz mid-band capacity second, and use 28 GHz only for targeted hotspots third.

1. Requirements and assumptions

The operator needs a network plan that serves 1,000 subscribers in the planning area at 0.08 Erlang each in the busy hour, with blocking of at most 2 %, a hexagonal layout of 19 cells of radius 2 km with reuse N = 7, support for mobiles at 60 km/h, and an interference level giving SINR of at least 15 dB. The report is organised as requirements → analysis → calculations → comparison → recommendation.

Global assumptions (additional ones are stated in each section):

- Free-space propagation is used only for the band comparison; real links include shadowing, building loss and clutter.
- Subscriber calls are independent, arrival is Poisson and blocked calls are cleared (Erlang B).
- Thermal noise is not given, so the "SINR" in Stage 5 contains interference only and is effectively an SIR. Including noise would lower it further.
- The speed of light $c = 3 \times 10^8$ m/s is used for all wavelength calculations.

2. Stage 1 – Evolution of mobile systems

2.1 Generational comparison

Gen.	Example / spectrum	Access / radio technology	Main service capability	Key limitation → motivation for next generation
1G	AMPS, 800 MHz, 30 kHz channels	Analog FM, FDMA	Analog voice only	Poor spectral efficiency, no encryption, no data, large reuse distance → digital transmission
2G	GSM, 900 MHz, 200 kHz carrier, 8 slots	Digital, TDMA/FDMA	Digital voice, SMS, low-rate data, encryption	Circuit-switched, very low data rates → higher-speed packet data
3G	UMTS, 5 MHz carrier	Wideband CDMA	Voice plus mobile data and multimedia	Interference-limited, data rate and latency limits → OFDM and more antennas
4G	LTE	OFDMA (downlink), MIMO, all-IP	Mobile broadband, video streaming	Not designed for ultra-low latency or massive device counts → service diversity
5G	NR: mid-band and mmWave	Flexible OFDM numerology, massive MIMO, beamforming	eMBB, URLLC, massive IoT	mmWave blockage and dense-site cost; network not natively sensing or AI-driven → 6G vision
6G	THz and new bands	AI-native networking, ISAC (integrated sensing and communication)	Holographic communication, tactile internet, sensing	Not yet standardised; THz range is very short and energy cost is a concern

Table 1. Generational comparison using the examples supplied in the planning data.

2.2 Calculations for 1G and the move to 2G

$$\lambda = c / f = (3 \times 10^8 \text{ m/s}) / (800 \times 10^6 \text{ Hz}) = 0.375 \text{ m}$$

$$N_ch = 10 \text{ MHz} / 30 \text{ kHz} = 333.3 \rightarrow 333 \text{ channels (idealised)}$$

Practical qualification. The 333 figure ignores guard bands, control channels and duplexing. A voice call needs a forward and a reverse channel (FDD), so if the 10 MHz is shared between both directions only about 166 duplex pairs would result, and some of those carry control signalling. The idealised number is therefore an upper bound, not usable traffic capacity.

1G versus 2G. Taking the same 10 MHz, GSM would give 10 MHz / 200 kHz = 50 carriers × 8 slots = 400 raw traffic channels (also before guard bands and control overhead), compared with 333 for AMPS. The larger gain is not the raw count. Digital transmission allows error-correcting codes, speech compression and equalisation, so GSM tolerates a lower carrier-to-interference ratio. Textbooks such as Rappaport typically quote about 18 dB for AMPS against about 9 dB for GSM, which allows tighter frequency reuse and so more capacity per area. TDMA also lets eight users share one carrier, which reduces the number of radio transceivers per cell. Digital service additionally brought encryption, SMS and better call quality.

2.3 Synthesis

Each generation changed how scarce spectrum is shared and how interference is controlled. 1G divided spectrum into fixed narrow channels, so capacity was bought mainly by cell splitting and strict reuse planning. 2G used digital coding and time slots to pack more users per hertz and reduce the required C/I. 3G reused the same wide carrier in every cell, turning interference into a power-control problem. 4G adopted OFDMA, which keeps users orthogonal within a cell, and MIMO, which adds spatial capacity, so interference management moved to inter-cell coordination and scheduling. 5G adds wide bandwidth in mid-band and mmWave plus beamforming, so capacity now depends on spectrum band choice, antenna arrays and network densification together. The 6G vision extends this

to THz bands and joint sensing and communication, with AI managing the network. Across the generations the pattern is the same: spectrum has moved from being divided into fixed channels to being managed dynamically in frequency, time, space and power.

3. Stage 2 – Cell coverage and frequency reuse

3.1 Calculations

$$A_cell = (3\sqrt{3} / 2)\, r^2 = 2.598 \times (2\ km)^2 = 10.39\ km^2$$

$$A_cluster(19\ cells) = 19 \times 10.39\ km^2 = 197.45\ km^2$$

$$Reuse\ fraction = 1/N = 1/7 \approx 0.143\ (14.3\ \%\ of\ the\ channels\ per\ cell)$$

$$D = r\sqrt{(3N)} = 2\ km \times \sqrt{21} = 9.17\ km, \quad D/r = 4.58$$

Quantity	Value	Unit
Cell radius r	2	km
Area of one hexagonal cell	10.39	km²
Area of 19-cell cluster	197.45	km²
Reuse cluster size N	7	–
Reuse fraction 1/N	0.143	–
Reuse distance D	9.17	km

Table 2. Geometry and reuse results.

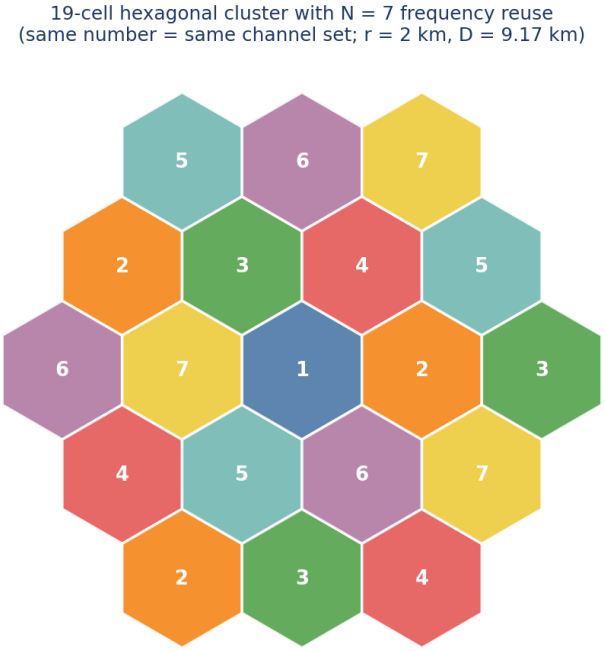


Figure 1. 19-cell hexagonal layout with N = 7 reuse colouring.

3.2 Interpretation and limits of the hexagonal model

The hexagon is a convenient approximation of a circular coverage area that tiles the plane without gaps or overlaps and gives a clean formula for reuse distance. It assumes a flat, uniform terrain, an omnidirectional antenna, equal base-station power, and uniform traffic. In practice terrain, buildings, antenna patterns and site availability make real cells irregular.

Geometric coverage versus guaranteed radio coverage. The 197 km² figure is only a geometric footprint. Radio coverage depends on received signal strength, fading margin, building penetration, noise and interference. Cell-edge users in a shadowed area may be inside the hexagon but still have no usable service, so coverage must be verified with link budgets, propagation prediction and drive tests.

3.3 Capacity-expansion measures

Measure	Effect on coverage and capacity	Main side effects
Cell splitting	Smaller radius with lower power; more cells per area, so more channel reuse and more Erlangs per km²	More sites and backhaul; more handoffs; reuse ratio D/R must be kept
Sectoring (e.g. 120°)	Directional antennas cut the number of first-tier co-channel interferers from 6 to about 2 (≈ 4.8 dB improvement in the ideal case) or allow smaller N	Channels per sector reduced (trunking loss); more handoffs between sectors
Microcells	Small outdoor cells in dense street areas add capacity where traffic is concentrated	Needs careful placement; more handoffs for fast users
Picocells	Indoor or hotspot cells (tens of metres) for office, mall, station	Needs backhaul and indoor planning
Femtocells	User-installed very small cells improve indoor coverage and offload macro cells	Uncoordinated deployment can create interference

Table 3. Capacity-expansion options.

Recommendation. For this operator the best first step is **120° sectorization of the existing macro sites, followed by selective cell splitting and microcells only in the busiest areas**. Sectorization reuses existing towers, directly improves the interference problem found in Stage 5, and requires no new spectrum. Its effects are: lower co-channel interference (higher SIR), a small trunking-efficiency loss because channels are divided among sectors, and more handoffs (inter-sector handoffs at the same site are cheaper than inter-site handoffs). Cell splitting later would increase handoff frequency roughly in proportion to the reduction in cell diameter and would also need re-planning of the reuse pattern.

4. Stage 3 – Traffic capacity

4.1 Offered traffic

$$A = N_sub \times a = 1{,}000 \times 0.08\ Erlang = 80\ Erlang$$

Assumptions: subscribers behave independently with the same average busy-hour load, call arrivals are Poisson, holding times are random, blocked calls are cleared, and the 1,000 subscribers are one pooled group served by a single channel group.

4.2 Erlang B model

The Erlang B recursion was evaluated in a short script (initial value B(0) = 1, then B(k) = A·B(k−1) / (k + A·B(k−1)) up to k = C). Inputs were A = 80 Erlang and C channels.

Channels C	Blocking P_B	Meets 2 % target?
90	2.62 %	No
91	2.25 %	No
92	1.92 %	Yes – smallest value
95	1.14 %	Yes
100	0.40 %	Yes (large margin)

Table 4. Erlang B blocking for A = 80 Erlang.

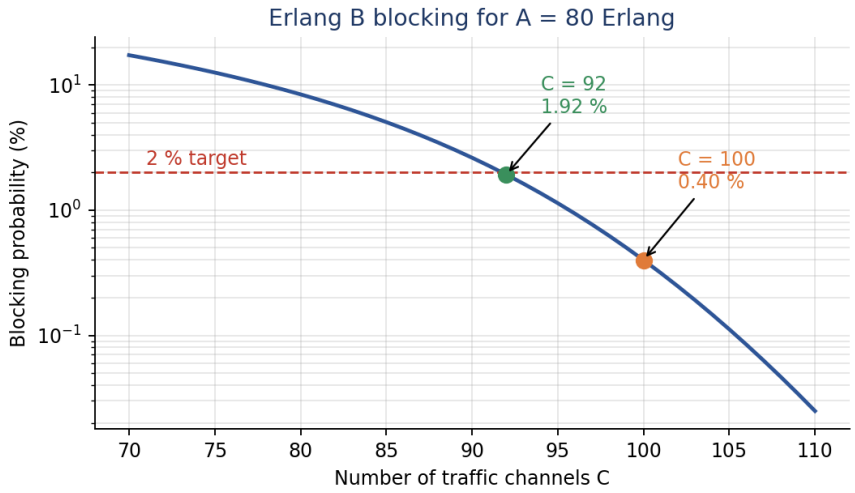


Figure 2. Blocking probability versus number of channels (A = 80 Erlang).

Results. With C = 100, P_B = 0.399 % (about 0.40 %), well below the target. The smallest channel count with P_B ≤ 2 % is **C = 92**. Equivalently, 100 channels can carry about 88 Erlang at 2 % blocking, about 1,100 subscribers at 0.08 Erlang.

Number of 100-channel groups. 92 / 100 → **1 group** ($\lceil 0.92 \rceil = 1$). One base-station channel group has about 8 spare channels.

4.3 Why this is only a first-order estimate

The pooled calculation assumes all 80 Erlang arrive at one place. In reality traffic is spread across 19 cells. If it were spread evenly, each cell would offer 80 / 19 = 4.21 Erlang and, by Erlang B, would need **9 channels** (1.72 % blocking; 8 channels gives 3.74 %). That is 19 × 9 = **171 channels** in total, almost twice the pooled value, because small groups of channels are less efficient than one large group (trunking efficiency). In addition, one site with r = 2 km covers only 10.39 km² of the 197 km² area, so coverage, not just capacity, requires several sites. A complete site plan would also need nonuniform traffic, hotspots, control channels, handoff reserve channels, sectoring, frequency reuse assignment, backhaul and growth margin.

Trade-offs. More channels reduce blocking but cost spectrum and radio equipment. Allowing 2 % instead of 0.4 % blocking saves about 8 channels per 100, a useful efficiency gain. Spreading traffic over more cells increases total channel count but also increases coverage and gives each user a better signal.

5. Stage 4 – Mobility and handoff

5.1 Calculations

$$v = 60\ km/h = 16.67\ m/s$$

$$Distance\ in\ overlap\ zone = v \times t = 16.67\ m/s \times 6\ s = 100\ m$$

$$Time\ per\ cell = 1{,}000\ m / 16.67\ m/s = 60\ s \rightarrow handoff\ rate = 1\ per\ minute$$

Interpretation of cell length. The 1 km "nominal cell length along the route" is taken as the distance the mobile travels between consecutive handoff points. Each such length therefore produces one handoff, giving about **1 hard handoff per minute**. (A route through the middle of a 2 km-radius cell would be longer, about 4 km, giving about 0.25 handoff/min, so 1/min is a conservative, high-mobility case.) Within the 6 s overlap zone, the 100 m region defines the window in which the network must complete the handoff.

5.2 Hard, soft and softer handoff

Type	Connection continuity	Resource use
Hard	Break-before-make: old link released before the new one is set; short interruption (here 0.5 s) and risk of drop	Uses one link at a time; simplest, lowest resource use
Soft	Make-before-break between different base stations (typical for CDMA); no interruption	Two or more links simultaneously, using more channel and network resources
Softer	Make-before-break between sectors of the same base station; combined at the site	Extra radio resource at one site; no extra backhaul to other sites

Table 5. Handoff types.

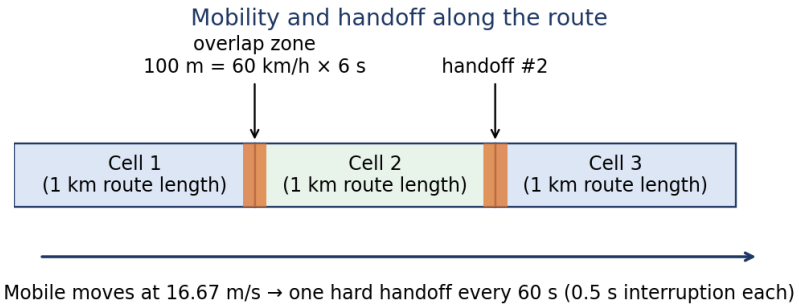


Figure 3. Cell, overlap zone and handoff spacing along the 60 km/h route.

5.3 Simple call-drop estimate

The target cell is unavailable with probability 0.01 at the instant of handoff. If a single attempt is made, the **call-drop risk per handoff is about P_drop ≈ 0.01 (1 %)**. At one handoff per minute this gives a drop probability of about 1 % per minute, or about $1 - 0.99^{10} \approx$ **9.6 % for a 10-minute call**. The 0.5 s interruption equals 0.5 / 60 = 0.83 % of the time at this handoff rate, and 8.3 % of the 6 s overlap window, so there is time for a retry.

Limits of the estimate. It treats handoffs as independent single attempts and fixed availability, ignoring retries within the overlap window (which would reduce the risk), correlated failures (blocking is worse in busy cells, which would increase it), signal fading, handoff-margin and hysteresis settings, reserved handoff channels and speed variation. It is an order-of-magnitude estimate only.

6. Stage 5 – Interference and spectrum bands

6.1 Interference calculation

Given S = −70 dBm, I_1 = −85 dBm (co-channel) and I_2 = −90 dBm (adjacent-channel).

$$C/I \text{ (co-channel only)} = S - I_1 = -70 - (-85) = 15 \text{ dB}$$

Interference powers must be added in linear units:

$$I_1 = 10^{\wedge}(-8.5)\ \text{mW} = 3.162 \times 10^{-9}\ \text{mW};\quad I_2 = 10^{\wedge}(-9.0)\ \text{mW} = 1.000 \times 10^{-9}\ \text{mW}$$

$$I_{total} = 4.162 \times 10^{-9}\ \text{mW} = 10 \log_{10}(4.162 \times 10^{-9}) = -83.81\ \text{dBm}$$

$$SINR = S - I_{total} = -70 - (-83.81) = 13.81\ \text{dB}$$

Quantity	Value	Meets 15 dB?
C/I, co-channel only	15.00 dB	Just meets (zero margin)
Total interference	−83.81 dBm	–
SINR with adjacent channel included	13.81 dB	No – short by 1.19 dB

Table 6. Interference results (noise not supplied, so excluded).

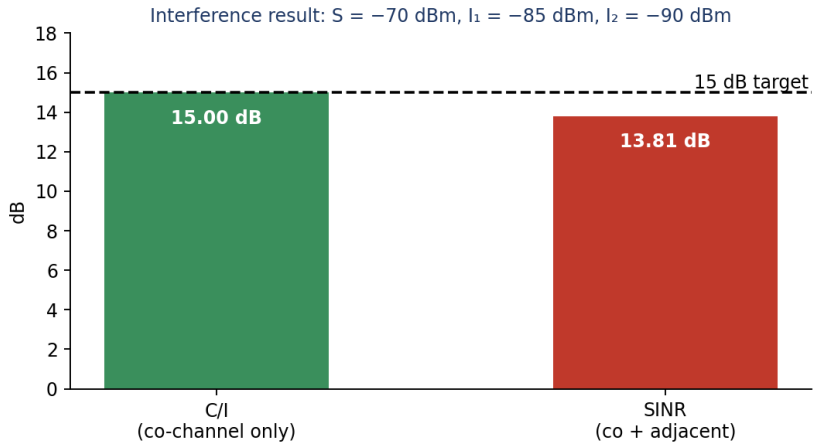


Figure 4. Co-channel C/I versus total SINR compared with the 15 dB target.

Decision. The 15 dB target is **not met**. The co-channel interferer alone leaves no margin and the adjacent-channel contribution adds about 1.2 dB of degradation. Real systems also need a fading margin, so the shortfall in practice would be larger.

Mitigation: 120° sectorization (with antenna downtilt). It reduces the number of co-channel interferers seen at a site from about six to about two, an ideal improvement of $10 \log_{10}(6/2) \approx 4.8$ dB, which exceeds the required 1.2 dB. Alternatives are increasing N (for example N = 12 gives D/r = 6.0) or improving receiver filtering and guard spacing against adjacent-channel interference. **Trade-off of sectorization:** each sector has fewer channels (trunking loss), more antennas and equipment are needed, and handoffs between sectors become more frequent. Increasing N would cost much more capacity per cell because each cell would then use only 1/12 of the channels.

6.2 Comparison of 900 MHz, 3.5 GHz and 28 GHz

$$\lambda = c / f; \quad FSPL(dB) = 20 \log_{10}(4\pi d / \lambda), \quad d = 100 \text{ m}$$

Band	Wavelength λ (m)	FSPL at 100 m (dB)	Extra loss vs 900 MHz (dB)	Ratio of λ to 900 MHz
900 MHz	0.3333 (33.3 cm)	71.53	0	1
3.5 GHz	0.0857 (8.6 cm)	83.32	+11.80	0.257
28 GHz	0.0107 (1.07 cm)	101.38	+29.86	0.032

Table 7. Wavelength and free-space path loss at 100 m. The extra loss equals $20 \log_{10}(f/900\ MHz)$.

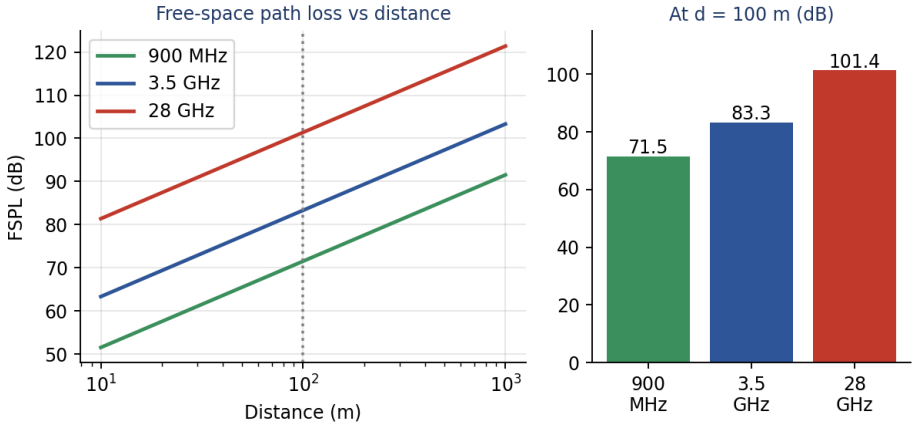


Figure 5. Free-space path loss of the three bands versus distance and at 100 m.

Band	Penetration and antenna size	Coverage	Capacity
900 MHz	Good building penetration; larger antennas (half-wave about 17 cm)	Largest cells; best for wide-area and rural coverage and indoor reach	Limited bandwidth, so lower peak capacity
3.5 GHz	Moderate penetration; compact arrays make massive MIMO practical	Medium cells; needs more sites than 900 MHz	Good balance; typically about 100 MHz channels
28 GHz	Poor penetration, strong blockage by walls, foliage and body; very small antenna elements, so large arrays in small area	Short range, line-of-sight; small cells only	Very wide bandwidth (hundreds of MHz), highest peak capacity per site in hotspots

Table 8. Qualitative deployment comparison.

Why path loss alone does not decide performance. Free-space loss at a fixed distance rises with frequency partly because the receiving antenna aperture shrinks with λ^2. With the same physical antenna area, higher-frequency arrays have more elements and higher gain, so beamforming can recover much of that loss. Network performance also depends on bandwidth (Shannon capacity grows linearly with bandwidth), transmit power, antenna gain, noise, interference, reuse, cell density and user distribution. A 28 GHz link may be far worse than 900 MHz in coverage yet far better in peak data rate where line of sight and small cells are available.

7. Stage 6 – Spectrum coordination and deployment recommendation

7.1 Roles in international spectrum coordination

Body	Role
ITU-R	The Radiocommunication Sector of the International Telecommunication Union. It develops recommendations and technical studies (for example the IMT-2020 and IMT-2030 frameworks) and supports the global management of radio-frequency spectrum and satellite orbits.
WRC	The World Radiocommunication Conference, held roughly every three to four years. It reviews and revises the Radio Regulations, the international treaty that allocates frequency bands to services and sets related rules, including identification of bands for IMT.
BR	The Radiocommunication Bureau, the executive arm of ITU-R. It processes and records frequency assignments and satellite filings, helps resolve interference, and supports the conferences and study groups.

Table 9. Roles of ITU-R, WRC and BR.

Effect on national regulators and operators. WRC decisions on band allocations and identification for IMT (for example for 3.5 GHz and millimetre-wave bands) guide national regulators when they define licences and auction or assign spectrum. Operators benefit because harmonised bands give larger device ecosystems and cheaper equipment, and because cross-border interference is limited by coordination. Operators can only deploy where their national regulator has licensed the band, so international decisions bring both opportunity and constraints (such as power limits and border coordination).

7.2 Recommended staged deployment strategy

Phase	Actions	Basis in the analysis
Phase 1 – Fix coverage and interference (0–12 months)	Deploy the 19-cell layout on sub-GHz (e.g. 900 MHz) spectrum, with 120° sectors and downtilt; plan for about 9 channels per cell (171 total) plus control and handoff reserve; tune handoff margins	900 MHz has the lowest path loss (71.5 dB at 100 m); sectoring corrects the 13.81 dB SINR failure; the per-cell Erlang B result sets channels; handoff about 1/min and 1 % drop risk need tuned thresholds and reserved channels

Phase	Actions	Basis in the analysis
Phase 2 – Add capacity (12–30 months)	Add 3.5 GHz mid-band carriers and massive MIMO on busy sites; microcells in dense corridors; consider cell splitting where traffic exceeds the per-cell plan	3.5 GHz costs +11.8 dB over 900 MHz but offers much more bandwidth; its small wavelength allows compact arrays; splitting increases reuse and capacity
Phase 3 – Hotspots and future services (30 months+)	Add 28 GHz small cells and pico/femto cells at stations, campuses and malls; prepare for URLLC and ISAC trials; follow WRC and ITU-R IMT-2030 work	28 GHz has +29.9 dB extra loss and poor penetration so only suitable for short-range hotspots; later generations need the new bands that WRC identifies

Table 10. Phased plan linking evolution, cell and traffic plan, handoff, interference and spectrum choices.

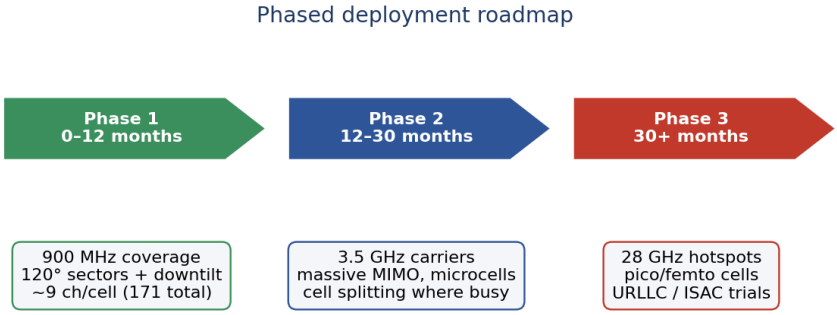


Figure 6. Three-phase deployment roadmap.

Evolution link. The phases follow the historical pattern: 1G/2G-style coverage first, 4G-style capacity growth with OFDMA/MIMO second, and 5G-style heterogeneous deployment with beamforming and eMBB/URLLC/IoT services third, leaving room for the 6G vision.

Trade-offs. Lower bands give coverage but little bandwidth; higher bands give capacity but need many more sites, backhaul and energy. Densification and sectoring improve capacity but increase handoff rate and cost. Pooling traffic reduces channel need but is unrealistic across 19 cells.

7.3 Assumptions and uncertainties to check before deployment

- Real traffic distribution and growth (the uniform 4.21 Erlang per cell and 0.08 Erlang per subscriber are assumptions); hotspots may need more than 9 channels.
- Actual propagation: path loss exponents, shadowing, indoor penetration and clutter, using drive tests and prediction tools rather than free-space loss.
- Interference: noise floor, real adjacent-channel leakage, antenna patterns and the actual SINR requirement of the radio technology used.
- Handoff behaviour: real route geometry (1 km cell length), overlap size, speed distribution, handoff thresholds and target-cell availability.
- Regulatory: which bands (900 MHz, 3.5 GHz, 28 GHz) the national regulator will license, with what bandwidth, power limits and border coordination requirements.
- Economics: site acquisition, backhaul, equipment and device availability for each band.

References

[1] T. S. Rappaport, Wireless Communications: Principles and Practice, 2nd ed., Prentice Hall, 2002 (cellular concepts, Erlang B, reuse, handoff, interference).

[2] ITU-R Recommendation M.2083-0, IMT Vision – Framework and overall objectives of the future development of IMT for 2020 and beyond, 2015.

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[4] ITU, Radio Regulations (edition in force) and information on the WRC and the Radiocommunication Bureau, www.itu.int/en/ITU-R.

[5] 3GPP specifications for LTE and 5G NR, www.3gpp.org.